



STILVerify® User's Manual

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Chapter 1

Getting Started

STILVerify™ is a free verification utility provided by Mentor Graphics for checking the conformity of Standard Test Interface Language (STIL) files. This tool enables developers of Electronic Design Automation (EDA), Automated Test Equipment (ATE), and related tools to ensure compatibility with STIL-based flows. STILVerify ensures that STIL files are syntactically correct, and it also features a Verilog testbench that allows EDA and ATE tool developers to run and display STIL content in any Verilog simulator in order to verify the content and behavior of STIL code. This feature also allows test engineers to define STIL-based flows more easily, enabling faster initial ramp up of STIL-based test processes. STILVerify reduces the time, effort, and expense required for semiconductor manufacturers to ramp up their STIL-based flows.

For the complete list of Mentor Graphics Tessent-specific terms, refer to the Tessent Glossary.

STILVerify is available for download at the following URL:

www.mentor.com/products/dft

STIL Language

STIL is an industry standard, IEEE-approved language providing an interface between digital test generation and ATE tools; and also provides data exchange between EDA point tools from different vendors. STIL is designed to transfer high density digital test patterns and waveforms between simulation, Automatic Test Pattern Generation (ATPG), integral test techniques such as Built-In-Self-Test (BIST), and ATE tools. STIL can be used to represent test patterns for both simple and complex devices.

Established by the IEEE in 1999 (the IEEE 1450 Standard), STIL is a test data description language designed to provide an unambiguous and complete standard data interface between EDA software and ATE interfaces, thereby simplifying semiconductor test flow. STIL is an open source language due to the requirement for the language to provide transportability from design contexts (EDA) to the test environment (ATE).

Converting EDA test patterns to proprietary ATE test languages is time consuming using manual translation methods. With a STIL based flow, test patterns can be generated for the ATE in less time, with less a likelihood for errors being generated during language translation. Note that STIL programs are representations of information needed to define digital test operations in manufacturing tests, and are not intended to define how the tester implements that information. STIL facilitates the transfer of digital test pattern data from EDA to ATE environments by supporting the volume of digital test pattern data generated by ATPG and

BIST tools, in a format optimized for application in ATE environments. The language allows for the specification of pattern, format, and timing information for defining the application of digital test patterns to a device to be tested. The result is a decrease in the time and expense of test program development, and an increase in the efficiency of running design-specific tests on target ATE machines.

STIL Standard

The emerging IEEE 1450 STIL standard includes several extensions to handle semiconductor design environments, DC-level specification, tester target and flows, test methods, and semiconductor cores. Some extensions have been adopted; others are in varying stages of development. In this document, when STIL is mentioned without a version number, it refers to the original STIL effort (now known as IEEE Std. 1450.0-1999), which defines the standard data format used to convey test pattern and timing information from ATPG tools to ATE test software generation tools.

The versions of the STIL Standard are as follows:

- **1450.0-1999** — Basic STIL for digital test patterns. This standard was approved by the IEEE in 1999 and is now known as IEEE Std. 1450.0-1999. As additional test areas were identified, follow-on extensions were defined. These are documented in the following items.
- **1450.1** — Extension to STIL for semiconductor design environments. This extension supports EDA-specific application of STIL data, test development, and test refinement processes.
- **1450.2-2002** — Extension to STIL for DC-level specification. This extension defines structures in STIL for specifying the dc conditions for a device under test. These conditions may be specified globally or by pattern burst, pattern, or vector. This standard was approved by the IEEE in 2002 and is now known as IEEE Std. 1450.2-2002.
- **1450.3** — Extension to STIL for tester resource constraints. This extension defines mechanisms for identifying constraints on STIL constructs in particular test environments.
- **1450.4** — Extension to STIL for test flow specification. This extension defines test organization, control, sequencing, and error processing.
- **1450.5** — Extension to STIL for semiconductor test method specification. This extension defines constructs for performing specific test operations.
- **1450.6** — Extension to STIL for Core Test Language (CTL) support. As part of the STIL standard, the IEEE created CTL, which is a standard description language that

gives designers a standard approach for describing DFT structures in core-based System-on-chip (SoC) designs when generating automatic test programs.

- **1450.AMS** — Extension to STIL for analog and mixed signal circuits or designs.

For more information on the STIL format, refer to IEEE Standard Test Interface Language (STIL) for Digital Test Vector Data, IEEE Std. 1450-1999.

STILVerify Overview

The STILVerify utility is used to parse STIL files from any tool and report warnings and errors for any problems found. It can also be used to create a standalone Verilog testbench in order to check the functionality of the STIL pattern data. STILVerify is a standalone command line utility with no GUI and no license requirements. The tool has the ability to create either serial or parallel Verilog testbenches for all STIL pattern files.

Note



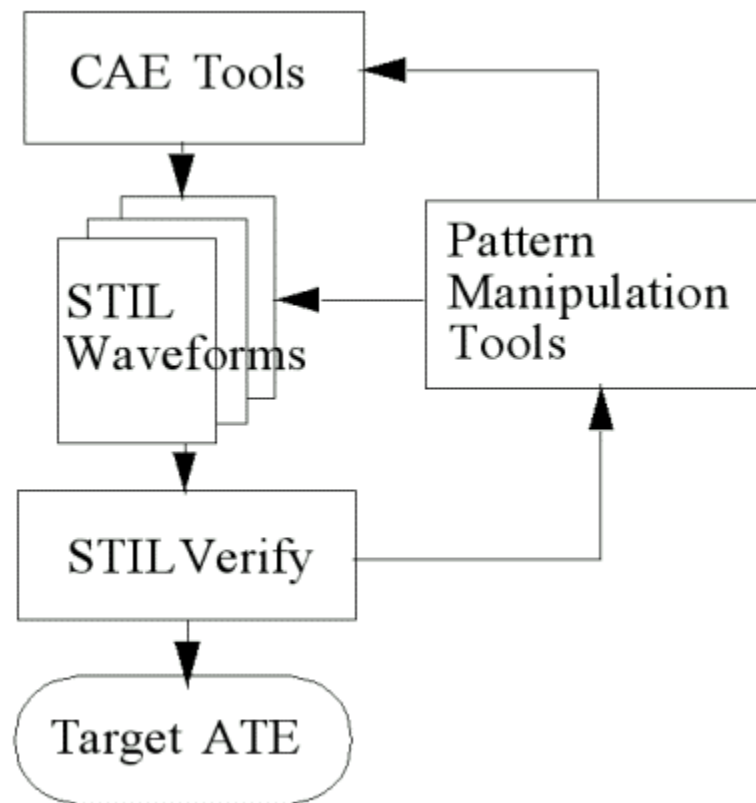
The STILVerify tool does not alter the original STIL file when it verifies its syntax.

The STILVerify utility contains an option to set a STIL compliance level in order to generate warnings if the STIL file uses STIL syntax that is not supported in the STIL “dot” extension specified. The STILVerify utility supports the following:

- Full support and checking for IEEE STIL 1450.0.
- Partial compliance for IEEE 1450.1, in that STILVerify can read “common” scan structures and generate parallel testbenches.
- Full compliance for IEEE STIL 1450.2.
- Partial support for IEEE 1450.6.

The following figure shows how STILVerify fits into the data flow between CAE and ATE environments. STIL represents patterns as a series of cyclized waveforms that are executed sequentially. The waveform representation can be simple or a complex set of parameterized events, so you might need tools to manipulate the STIL data according to the requirements of the ATE device, though the output of that manipulation will still be represented in STIL.

Figure 1-1. STIL Data Flow



Using STILVerify

This section contains a brief guide to usage and introductory exercises you can run. It is intended to familiarize you with the operation of the STILVerify tool. This section does not provide full details for running the tool, but rather, contains enough information to help you get started.

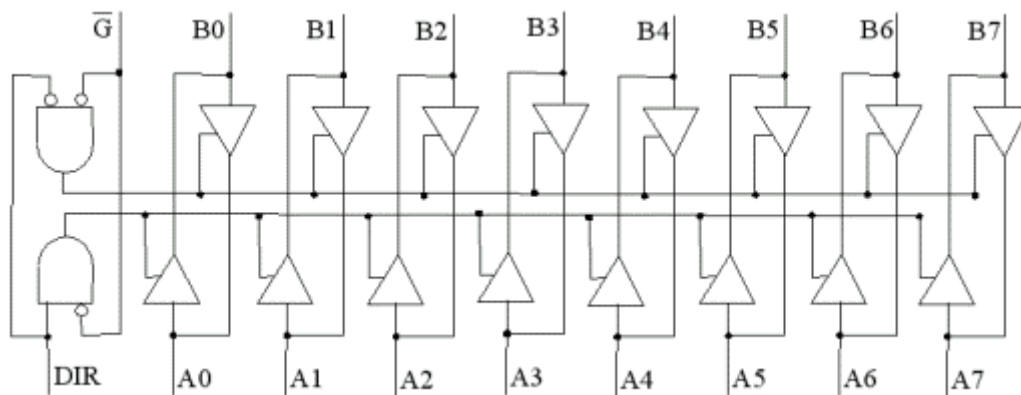
In order to use the STILVerify utility, you will need some source for your STIL patterns, such as a tool, or the ability to create them manually. See [stilverify](#) for a description of all STILVerify command arguments

STILVerify Procedure

This describes the procedure for using STILVerify.

This description for STILVerify uses a 74245 8-bit tri-state noninverting bus transceiver, shown in [Figure 1-2](#). Enable and direction pins control output enables. The circuit has been simplified for this example so that bus signals are unidirectional.

Figure 1-2. Block Diagram of 8-Bit Bus Transceiver



Prerequisites

- A complete STIL program. For example, this file, *test.stil*, represents a complete STIL program to exercise an 8-bit bus transceiver:

```
// STIL example of an 8-bit bus transceiver
STIL 1.0;
// The Header block specifies data about the STIL file.
Header {
    Title "example stil file -- stest.stil";
    Date "Fri Jan 12 11:33:54 2006";
    Source "Simplified 74245";
    History { Ann { * rev 1 -- original version * } }
}
// The Signals block defines individual signal names used
// in the design.
Signals {
    G_ In; DIR In;
    A0 In; A1 In; A2 In; A3 In; A4 In; A5 In; A6 In; A7 In;
    B0 Out; B1 Out; B2 Out; B3 Out; B4 Out; B5 Out; B6 Out; B7 Out;
}
// The SignalGroups block defines named references to signals.
SignalGroups {
    A = 'A7 + A6 + A5 + A4 + A3 + A2 + A1 + A0';
    B = 'B7 + B6 + B5 + B4 + B3 + B2 + B1 + B0';
    BUSES = 'DIR + G_ + A + B';
}
// The Timing block defines a sequence of repeated waveforms
// to execute test programs.
Timing normal {
    WaveformTable defaults {
        Period '500ns';
        Waveforms {
            DIR { 01 { '0ns' D/U; } }
            G_ { 01 { '0ns' U; '200ns' D/U; '300ns' U; } }
            A { 01 { '10ns' D/U; } }
            B { HLZ { '0ns' Z; '0ns' X; '260ns' h/l/t; '280ns' X; } }
        }
    }
}
// The PatternBurst block defines pattern names to be executed.
PatternBurst all_pats { PatList { Pat1; } }

// The PatternExec block defines how Timing and PatternBurst
// information is assembled to create the tests to execute.
PatternExec run_all {
    Timing normal;
    PatternBurst all_pats;
}

// The Include statement defines a file reference.
Include "pat1.stil";
```

- A file containing the STIL test pattern data. For example this test pattern data file, *pat1.stil*:

```
// STIL test pattern data

// The Pattern block defines the operation of the test, by repeated
// application of waveforms for each signal.
Pattern Pat1 {
    WaveformTable defaults;
    Vector { BUSES = 0000000000LLLLLLLL; }
    Vector { BUSES = 0010000000HLLLLLLLL; }
    Vector { BUSES = 0001000000LHLLLLLLLL; }
    Vector { BUSES = 0000100000LLHLLLLLLL; }
    Vector { BUSES = 0000010000LLLHLLLLLL; }
    Vector { BUSES = 0000001000LLLLHLLLL; }
    Vector { BUSES = 0000000100LLLLLHLL; }
    Vector { BUSES = 0000000010LLLLLLHL; }
    Vector { BUSES = 0000000001LLLLLLLH; }
}
```

Procedure

1. Enter the stilverify command at the shell command prompt to invoke STILVerify. For example, the following command will parse the file *stest.stil*, and create a testbench *btest.v* with “btransceiver” as the top-level module name for this particular design.

```
stilverify stest.stil -verilog btest.v -serial \
-module btransceiver -log test.log -replace
```

2. After the tool finishes and exits, you may review the command results. In this case, the transcript was saved in the logfile named *test.log*. The logfile contains a copy of the command you executed, as well as the results of executing the command. In this case, you see the message “No errors found in STIL file” indicating there are no syntax errors in the STIL file:

```
stilverify stest.stil -verilog btest.v -serial -module btransceiver
\ -log test.log -replace
// Reading STIL input file "stest.stil"
// Finished parsing STIL input file "stest.stil"
// No errors found in STIL file.
// Creating Verilog testbench "btest.v"
```

3. Once you are sure the syntax in the STIL patterns is correct, you can use a digital simulator such as ModelSim® to run a simulation on the Verilog testbench that STILVerify generated to ensure that the content of the patterns is what you expected.

To be able to run the testbench, you need to have an RTL or gate-level Verilog description of the design as well as Verilog simulation libraries. If there are simulation mismatches, they will be transcribed to the output. Simulation mismatches indicate that your STIL patterns do not appear to match the design, and you will need to determine why.

A typical ModelSim run of the STIL Verify testbench looks like this:

```
vlog btest.v

Model Technology ModelSim SE vlog 5.8a compiler 2003.12 Dec 19 2003
-- Compiling module btransceiver_btest_v_ctl

Top-level modules:
    btransceiver_btest_v_ctl

vsim +notimingchecks -c btransceiver_btest_v_ctl
# // ModelSim SE 5.8a Dec 01 2003 SunOS 5.8
# //
# // Copyright Model Technology, a Mentor Graphics Corp. company,
2003
# //
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# // PROPERTY OF MENTOR GRAPHICS CORPORATION OR ITS LICENSORS.
# //
# Loading work.btransceiver_btest_v_ctl
# Loading work.btransceiver

... Loading statements and warnings about design being simulated ...

VSIM 1> run -all
# Loading btest.v.0.vec
#
# No error between simulated and expected patterns
#
# ** Note: $finish : ./btest.v(190)
```

stilverify

Verifies the correctness of STIL generated and used by ATE and EDA tools.

Usage

```
stilverify stil_pathname [-EXtension extension_number[-EXtension extension_number ...]][-  
    Verilog test_file] [-SErial] [-MOdule module_name] [-PARAMeter <file_name>][-  
    VERBose] [-LOgfile logfile [-REplace]] [-Help] [-Usage] [-VERSion]
```

Arguments

- **stil_pathname**

A required string pair that specifies the input STIL pattern file to check.

- **-EXtension number**

An optional switch and string pair that specifies a STIL dot extension to use for compliance checking (0 through 6). If specified, any STIL syntax found in the file that comes from extensions not specified will produce a warning. If not specified, all supported STIL levels are accepted without warning. This switch allows multiple values for number and multiple -Extension switches; therefore, to specify both STIL 1450.1 and 1450.2, either of the following is acceptable:

- -ex 1 2
- -ex 1 -ex 2

Note



Entering “-ex 0” is not necessary, as STIL 1450.0 is the basis for all other extensions, and it is impossible to not use the syntax in 1450.0.

- **-VERilog test_file**

An optional switch and string pair that specifies that a Verilog testbench should be written. If this argument is not specified, no testbench will be produced.

- **-SErial**

Optional switch that specifies whether to produce a serial scan load Verilog testbench. This is the default.

- **-MOdule module_name**

An optional switch and string pair that specifies the Verilog module name to use for the design being instantiated in the testbench. If not specified, the tool will attempt to create the module name from information in the STIL file.

Note



You can find the top-level module name from your design netlist.

- **-PARAMeter <file_name>**
An optional switch and string pair that specifies to load a parameter file which is used when writing a Verilog testbench if the **-VERilog** switch is used. All Verilog-related parameter file keywords from the ATPG tools are supported in this parameter file.
- **-VERbose**
An optional switch that prints more detailed warning messages concerned with STIL practices that are not recommended. The STIL code that will result in these warnings is still technically legal.
- **-LOGfile logfile**
An optional switch and string pair that specifies the logfile to write.
- **-REplace**
An optional switch that specifies to replace the logfile if it already exists.
- **-Help**
An optional switch that displays a message that contains all the STILVerify switches and a brief description of each.
- **-Usage**
An optional switch that displays the usage syntax for the STILVerify command.
- **-Version**
An optional switch that displays the version of the STILVerify tool that you currently have available.

Description

Verifies the correctness of STIL files generated by any tool and reports warnings and errors for any problems found. The tool can also be used to create a standalone, serial or parallel Verilog testbench, in order to check the functionality of the STIL pattern data.

The STILVerify command is entered on the command line in lowercase, as in “stilverify”. The uppercase characters on the arguments represent minimal parameter representation. Also, do not enter the square braces ([]) when typing the command. Use of brackets means that the argument is optional.

Examples

This command will parse the file *stest.stil*, and create a testbench *btest.v* with “btransceiver” as the top-level module name for this particular design.

```
stilverify stest.stil -verilog btest.v  
-serial -module btransceiver -log test.log -replace
```


There are several ways to get help when setting up and using Tessent software tools. Depending on your need, help is available from documentation, online command help, and Mentor Graphics Support.

Documentation

The Tessent software tree includes a complete set of documentation and help files in PDF format. Although you can view this documentation with any PDF reader, if you are viewing documentation on a Linux file server, you must use only Adobe® Reader® versions 8 or 9, and you must set one of these versions as the default using the MGC_PDF_READER variable in your *mgc_doc_options.ini* file.

For more information, refer to “[Specifying Documentation System Defaults](#)” in the *Managing Mentor Graphics Tessent Software* manual.

You can download a free copy of the latest Adobe Reader from this location:

get.adobe.com/reader

You can access the documentation in the following ways:

- **Shell Command** — On Linux platforms, enter **mgcdocs** at the shell prompt or invoke a Tessent tool with the -Manual invocation switch. This option is available only with Tessent Shell and the following classic point tools: Tessent FastScan, Tessent TestKompress, Tessent Diagnosis, and DFTAdvisor.
- **File System** — Access the Tessent bookcase directly from your file system, without invoking a Tessent tool. From your product installation, invoke Adobe Reader on the following file:

```
$MGC_DFT/doc/pdfdocs/_bk_tessent.pdf
```

- **Application Online Help** — You can get contextual online help within most Tessent tools by using the “help -manual” tool command:

```
> help dofile -manual
```

This command opens the appropriate reference manual at the “dofile” command description.

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Third-Party Information

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For third-party information, refer to the [*Third-Party Software for Tessent*](#) products.

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- 3.3. Customer agrees to maintain Beta Code in confidence and shall restrict access to the Beta Code, including the methods and concepts utilized therein, solely to those employees and Customer location(s) authorized by Mentor Graphics to perform beta testing. Customer agrees that any written evaluations and all inventions, product improvements, modifications or developments that Mentor Graphics conceived or made during or subsequent to this Agreement, including those based partly or wholly on Customer's feedback, will be the exclusive property of Mentor Graphics. Mentor Graphics will have exclusive rights, title and interest in all such property. The provisions of this Subsection 3.3 shall survive termination of this Agreement.

4. RESTRICTIONS ON USE.

- 4.1. Customer may copy Software only as reasonably necessary to support the authorized use. Each copy must include all notices and legends embedded in Software and affixed to its medium and container as received from Mentor Graphics. All copies shall remain the property of Mentor Graphics or its licensors. Except for Embedded Software that has been embedded in executable code form in Customer's product(s), Customer shall maintain a record of the number and primary location of all copies of Software, including copies merged with other software, and shall make those records available to Mentor Graphics upon request. Customer shall not make Products available in any form to any person other than Customer's employees and on-site contractors, excluding Mentor Graphics competitors, whose job performance requires access and who are under obligations of confidentiality. Customer shall take appropriate action to protect the confidentiality of Products and ensure that any person permitted access does not disclose or use Products except as permitted by this Agreement. Customer shall give Mentor Graphics written notice of any unauthorized disclosure or use of the Products as soon as Customer becomes aware of such unauthorized disclosure or use. Customer acknowledges that Software provided hereunder may contain source code which is proprietary and its confidentiality is of the highest importance and value to Mentor Graphics. Customer acknowledges that Mentor Graphics may be seriously harmed if such source code is disclosed in violation of this Agreement. Except as otherwise permitted for purposes of interoperability as specified by applicable and mandatory local law, Customer shall not reverse-assemble, disassemble, reverse-compile, or reverse-engineer any Product, or in any way derive any source code from Software that is not provided to Customer in source code form. Log files, data files, rule files and script files generated by or for the Software (collectively "Files"), including without limitation files containing Standard Verification Rule Format ("SVRF") and Tcl Verification Format ("TVF") which are Mentor Graphics' trade secret and proprietary syntaxes for expressing process rules, constitute or include confidential information of Mentor Graphics. Customer may share Files with third parties, excluding Mentor Graphics competitors, provided that the confidentiality of such Files is protected by written agreement at least as well as Customer protects other information of a similar nature or importance, but in any case with at least reasonable care. Customer may use Files containing SVRF or TVF only with Mentor Graphics products. Under no circumstances shall Customer use Products or Files or allow their use for the purpose of developing, enhancing or marketing any product that is in any way competitive with Products, or disclose to any third party the results of, or information pertaining to, any benchmark.
 - 4.2. If any Software or portions thereof are provided in source code form, Customer will use the source code only to correct software errors and enhance or modify the Software for the authorized use, or as permitted for Embedded Software under separate embedded software terms or an embedded software supplement. Customer shall not disclose or permit disclosure of source code, in whole or in part, including any of its methods or concepts, to anyone except Customer's employees or on-site contractors, excluding Mentor Graphics competitors, with a need to know. Customer shall not copy or compile source code in any manner except to support this authorized use.
 - 4.3. Customer agrees that it will not subject any Product to any open source software ("OSS") license that conflicts with this Agreement or that does not otherwise apply to such Product.
 - 4.4. Customer may not assign this Agreement or the rights and duties under it, or relocate, sublicense, or otherwise transfer the Products, whether by operation of law or otherwise ("Attempted Transfer"), without Mentor Graphics' prior written consent and payment of Mentor Graphics' then-current applicable relocation and/or transfer fees. Any Attempted Transfer without Mentor Graphics' prior written consent shall be a material breach of this Agreement and may, at Mentor Graphics' option, result in the immediate termination of the Agreement and/or the licenses granted under this Agreement. The terms of this Agreement, including without limitation the licensing and assignment provisions, shall be binding upon Customer's permitted successors in interest and assigns.
 - 4.5. The provisions of this Section 4 shall survive the termination of this Agreement.
5. **SUPPORT SERVICES.** To the extent Customer purchases support services, Mentor Graphics will provide Customer with updates and technical support for the Products, at the Customer site(s) for which support is purchased, in accordance with Mentor Graphics' then current End-User Support Terms located at <http://supportnet.mentor.com/supportterms>.
6. **OPEN SOURCE SOFTWARE.** Products may contain OSS or code distributed under a proprietary third party license agreement, to which additional rights or obligations ("Third Party Terms") may apply. Please see the applicable Product documentation (including license files, header files, read-me files or source code) for details. In the event of conflict between the terms of this Agreement

(including any addenda) and the Third Party Terms, the Third Party Terms will control solely with respect to the OSS or third party code. The provisions of this Section 6 shall survive the termination of this Agreement.

7. LIMITED WARRANTY.

- 7.1. Mentor Graphics warrants that during the warranty period its standard, generally supported Products, when properly installed, will substantially conform to the functional specifications set forth in the applicable user manual. Mentor Graphics does not warrant that Products will meet Customer's requirements or that operation of Products will be uninterrupted or error free. The warranty period is 90 days starting on the 15th day after delivery or upon installation, whichever first occurs. Customer must notify Mentor Graphics in writing of any nonconformity within the warranty period. For the avoidance of doubt, this warranty applies only to the initial shipment of Software under an Order and does not renew or reset, for example, with the delivery of (a) Software updates or (b) authorization codes or alternate Software under a transaction involving Software re-mix. This warranty shall not be valid if Products have been subject to misuse, unauthorized modification, improper installation or Customer is not in compliance with this Agreement. MENTOR GRAPHICS' ENTIRE LIABILITY AND CUSTOMER'S EXCLUSIVE REMEDY SHALL BE, AT MENTOR GRAPHICS' OPTION, EITHER (A) REFUND OF THE PRICE PAID UPON RETURN OF THE PRODUCTS TO MENTOR GRAPHICS OR (B) MODIFICATION OR REPLACEMENT OF THE PRODUCTS THAT DO NOT MEET THIS LIMITED WARRANTY. MENTOR GRAPHICS MAKES NO WARRANTIES WITH RESPECT TO: (A) SERVICES; (B) PRODUCTS PROVIDED AT NO CHARGE; OR (C) BETA CODE; ALL OF WHICH ARE PROVIDED "AS IS."
- 7.2. THE WARRANTIES SET FORTH IN THIS SECTION 7 ARE EXCLUSIVE. NEITHER MENTOR GRAPHICS NOR ITS LICENSORS MAKE ANY OTHER WARRANTIES EXPRESS, IMPLIED OR STATUTORY, WITH RESPECT TO PRODUCTS PROVIDED UNDER THIS AGREEMENT. MENTOR GRAPHICS AND ITS LICENSORS SPECIFICALLY DISCLAIM ALL IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT OF INTELLECTUAL PROPERTY.

8. **LIMITATION OF LIABILITY.** TO THE EXTENT PERMITTED UNDER APPLICABLE LAW, IN NO EVENT SHALL MENTOR GRAPHICS OR ITS LICENSORS BE LIABLE FOR INDIRECT, SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES (INCLUDING LOST PROFITS OR SAVINGS) WHETHER BASED ON CONTRACT, TORT OR ANY OTHER LEGAL THEORY, EVEN IF MENTOR GRAPHICS OR ITS LICENSORS HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. IN NO EVENT SHALL MENTOR GRAPHICS' OR ITS LICENSORS' LIABILITY UNDER THIS AGREEMENT EXCEED THE AMOUNT RECEIVED FROM CUSTOMER FOR THE HARDWARE, SOFTWARE LICENSE OR SERVICE GIVING RISE TO THE CLAIM. IN THE CASE WHERE NO AMOUNT WAS PAID, MENTOR GRAPHICS AND ITS LICENSORS SHALL HAVE NO LIABILITY FOR ANY DAMAGES WHATSOEVER. THE PROVISIONS OF THIS SECTION 8 SHALL SURVIVE THE TERMINATION OF THIS AGREEMENT.

9. THIRD PARTY CLAIMS.

- 9.1. Customer acknowledges that Mentor Graphics has no control over the testing of Customer's products, or the specific applications and use of Products. Mentor Graphics and its licensors shall not be liable for any claim or demand made against Customer by any third party, except to the extent such claim is covered under Section 10.
- 9.2. In the event that a third party makes a claim against Mentor Graphics arising out of the use of Customer's products, Mentor Graphics will give Customer prompt notice of such claim. At Customer's option and expense, Customer may take sole control of the defense and any settlement of such claim. Customer WILL reimburse and hold harmless Mentor Graphics for any LIABILITY, damages, settlement amounts, costs and expenses, including reasonable attorney's fees, incurred by or awarded against Mentor Graphics or its licensors in connection with such claims.
- 9.3. The provisions of this Section 9 shall survive any expiration or termination of this Agreement.

10. INFRINGEMENT.

- 10.1. Mentor Graphics will defend or settle, at its option and expense, any action brought against Customer in the United States, Canada, Japan, or member state of the European Union which alleges that any standard, generally supported Product acquired by Customer hereunder infringes a patent or copyright or misappropriates a trade secret in such jurisdiction. Mentor Graphics will pay costs and damages finally awarded against Customer that are attributable to such action. Customer understands and agrees that as conditions to Mentor Graphics' obligations under this section Customer must: (a) notify Mentor Graphics promptly in writing of the action; (b) provide Mentor Graphics all reasonable information and assistance to settle or defend the action; and (c) grant Mentor Graphics sole authority and control of the defense or settlement of the action.
- 10.2. If a claim is made under Subsection 10.1 Mentor Graphics may, at its option and expense: (a) replace or modify the Product so that it becomes noninfringing; (b) procure for Customer the right to continue using the Product; or (c) require the return of the Product and refund to Customer any purchase price or license fee paid, less a reasonable allowance for use.
- 10.3. Mentor Graphics has no liability to Customer if the action is based upon: (a) the combination of Software or hardware with any product not furnished by Mentor Graphics; (b) the modification of the Product other than by Mentor Graphics; (c) the use of other than a current unaltered release of Software; (d) the use of the Product as part of an infringing process; (e) a product that Customer makes, uses, or sells; (f) any Beta Code or Product provided at no charge; (g) any software provided by Mentor Graphics' licensors who do not provide such indemnification to Mentor Graphics' customers; (h) OSS, except to the extent that the infringement is directly caused by Mentor Graphics' modifications to such OSS; or (i) infringement by Customer that is deemed willful. In the case of (i), Customer shall reimburse Mentor Graphics for its reasonable attorney fees and other costs related to the action.
- 10.4. THIS SECTION 10 IS SUBJECT TO SECTION 8 ABOVE AND STATES THE ENTIRE LIABILITY OF MENTOR GRAPHICS AND ITS LICENSORS, AND CUSTOMER'S SOLE AND EXCLUSIVE REMEDY, FOR DEFENSE,

SETTLEMENT AND DAMAGES, WITH RESPECT TO ANY ALLEGED PATENT OR COPYRIGHT INFRINGEMENT OR TRADE SECRET MISAPPROPRIATION BY ANY PRODUCT PROVIDED UNDER THIS AGREEMENT.

11. TERMINATION AND EFFECT OF TERMINATION.

- 11.1. If a Software license was provided for limited term use, such license will automatically terminate at the end of the authorized term. Mentor Graphics may terminate this Agreement and/or any license granted under this Agreement immediately upon written notice if Customer: (a) exceeds the scope of the license or otherwise fails to comply with the licensing or confidentiality provisions of this Agreement, or (b) becomes insolvent, files a bankruptcy petition, institutes proceedings for liquidation or winding up or enters into an agreement to assign its assets for the benefit of creditors. For any other material breach of any provision of this Agreement, Mentor Graphics may terminate this Agreement and/or any license granted under this Agreement upon 30 days written notice if Customer fails to cure the breach within the 30 day notice period. Termination of this Agreement or any license granted hereunder will not affect Customer's obligation to pay for Products shipped or licenses granted prior to the termination, which amounts shall be payable immediately upon the date of termination.
- 11.2. Upon termination of this Agreement, the rights and obligations of the parties shall cease except as expressly set forth in this Agreement. Upon termination of this Agreement and/or any license granted under this Agreement, Customer shall ensure that all use of the affected Products ceases, and shall return hardware and either return to Mentor Graphics or destroy Software in Customer's possession, including all copies and documentation, and certify in writing to Mentor Graphics within ten business days of the termination date that Customer no longer possesses any of the affected Products or copies of Software in any form.
12. **EXPORT.** The Products provided hereunder are subject to regulation by local laws and European Union ("E.U.") and United States ("U.S.") government agencies, which prohibit export, re-export or diversion of certain products, information about the products, and direct or indirect products thereof, to certain countries and certain persons. Customer agrees that it will not export or re-export Products in any manner without first obtaining all necessary approval from appropriate local, E.U. and U.S. government agencies. If Customer wishes to disclose any information to Mentor Graphics that is subject to any E.U., U.S. or other applicable export restrictions, including without limitation the U.S. International Traffic in Arms Regulations (ITAR) or special controls under the Export Administration Regulations (EAR), Customer will notify Mentor Graphics personnel, in advance of each instance of disclosure, that such information is subject to such export restrictions.
13. **U.S. GOVERNMENT LICENSE RIGHTS.** Software was developed entirely at private expense. The parties agree that all Software is commercial computer software within the meaning of the applicable acquisition regulations. Accordingly, pursuant to U.S. FAR 48 CFR 12.212 and DFAR 48 CFR 227.7202, use, duplication and disclosure of the Software by or for the U.S. government or a U.S. government subcontractor is subject solely to the terms and conditions set forth in this Agreement, which shall supersede any conflicting terms or conditions in any government order document, except for provisions which are contrary to applicable mandatory federal laws.
14. **THIRD PARTY BENEFICIARY.** Mentor Graphics Corporation, Mentor Graphics (Ireland) Limited, Microsoft Corporation and other licensors may be third party beneficiaries of this Agreement with the right to enforce the obligations set forth herein.
15. **REVIEW OF LICENSE USAGE.** Customer will monitor the access to and use of Software. With prior written notice and during Customer's normal business hours, Mentor Graphics may engage an internationally recognized accounting firm to review Customer's software monitoring system and records deemed relevant by the internationally recognized accounting firm to confirm Customer's compliance with the terms of this Agreement or U.S. or other local export laws. Such review may include FlexNet (or successor product) report log files that Customer shall capture and provide at Mentor Graphics' request. Customer shall make records available in electronic format and shall fully cooperate with data gathering to support the license review. Mentor Graphics shall bear the expense of any such review unless a material non-compliance is revealed. Mentor Graphics shall treat as confidential information all information gained as a result of any request or review and shall only use or disclose such information as required by law or to enforce its rights under this Agreement. The provisions of this Section 15 shall survive the termination of this Agreement.
16. **CONTROLLING LAW, JURISDICTION AND DISPUTE RESOLUTION.** The owners of certain Mentor Graphics intellectual property licensed under this Agreement are located in Ireland and the U.S. To promote consistency around the world, disputes shall be resolved as follows: excluding conflict of laws rules, this Agreement shall be governed by and construed under the laws of the State of Oregon, U.S., if Customer is located in North or South America, and the laws of Ireland if Customer is located outside of North or South America or Japan, and the laws of Japan if Customer is located in Japan. All disputes arising out of or in relation to this Agreement shall be submitted to the exclusive jurisdiction of the courts of Portland, Oregon when the laws of Oregon apply, or Dublin, Ireland when the laws of Ireland apply, or the Tokyo District Court when the laws of Japan apply. Notwithstanding the foregoing, all disputes in Asia (excluding Japan) arising out of or in relation to this Agreement shall be resolved by arbitration in Singapore before a single arbitrator to be appointed by the chairman of the Singapore International Arbitration Centre ("SIAC") to be conducted in the English language, in accordance with the Arbitration Rules of the SIAC in effect at the time of the dispute, which rules are deemed to be incorporated by reference in this section. Nothing in this section shall restrict Mentor Graphics' right to bring an action (including for example a motion for injunctive relief) against Customer in the jurisdiction where Customer's place of business is located. The United Nations Convention on Contracts for the International Sale of Goods does not apply to this Agreement.
17. **SEVERABILITY.** If any provision of this Agreement is held by a court of competent jurisdiction to be void, invalid, unenforceable or illegal, such provision shall be severed from this Agreement and the remaining provisions will remain in full force and effect.
18. **MISCELLANEOUS.** This Agreement contains the parties' entire understanding relating to its subject matter and supersedes all prior or contemporaneous agreements. Any translation of this Agreement is provided to comply with local legal requirements only. In the event of a dispute between the English and any non-English versions, the English version of this Agreement shall govern to the extent not prohibited by local law in the applicable jurisdiction. This Agreement may only be modified in writing, signed by an authorized representative of each party. Waiver of terms or excuse of breach must be in writing and shall not constitute subsequent consent, waiver or excuse.